

MAE 4730
Lecture 36
11/19/18

Today:
· 3D dynamics
· $\underline{R}$, $\underline{I}$, $\underline{\omega}$

Recall:

LMB: $\sum \vec{F}^{ext} = \begin{cases} \sum m_i \vec{a}_i \\ = \int \vec{a} dm \\ = m_{tot} \vec{a}_G \end{cases}$

If only interested
relative to CoM

$$\vec{M}_{/G} = \int \vec{r}_{/G} \times \vec{a}_{/G} dm$$

AMB_{/c}: $\sum \vec{M}_{/c}^{ext} = \begin{cases} \sum \vec{r}_{i/c} \times m_i \vec{a}_{i/f} \\ \int \vec{r} \times \vec{a}_{/f} dm \\ \vec{r}_{G/c} \times m_{tot} \vec{a}_{G/f} + \int \vec{r}_{/G} \times \vec{a}_{/G} dm \end{cases}$

$\vec{a}_{/f} - \vec{a}_{G/f}$

" $\dot{\vec{H}}_{/G}$ "

$$\int \vec{r}_{/G} \times \vec{a}_{/G} dm = \frac{d}{dt} \left[\underbrace{\int \vec{r}_{/G} \times \vec{v}_{/G} dm}_{\vec{H}_{/G}} \right]$$

What is $\vec{H}_{/G}$ for a rigid object?

 $\vec{H}_{/G} = \int \vec{r}_{/G} \times \vec{v}_{/G} dm$

$\downarrow \vec{\omega} \times \vec{r}_{/G}$

$\downarrow \vec{\omega}_{P/f}$

$$= \int \vec{r}_{/G} \times (\vec{\omega} \times \vec{r}_{/G}) dm$$

$$\vec{a} \times (\vec{b} \times \vec{c}) = \vec{b}(\vec{a} \cdot \vec{c}) - \vec{c}(\vec{a} \cdot \vec{b})$$

$$\begin{aligned} \Rightarrow \vec{r} \times (\vec{\omega} \times \vec{r}) &= \vec{\omega}(\vec{r} \cdot \vec{r}) - \vec{r}(\vec{\omega} \cdot \vec{r}) \\ &= (\vec{r} \cdot \vec{r}) \vec{\omega} - \vec{r}(\vec{r} \cdot \vec{\omega}) \\ &= ((\vec{r} \cdot \vec{r}) \underline{1} - \vec{r}\vec{r}) \cdot \vec{\omega} \end{aligned}$$

$$\Rightarrow \vec{H}_{/G} = \left[\int (\vec{r} \cdot \vec{r}) dm \underline{1} - \left[\int \vec{r} \vec{r} dm \right] \right] \cdot \vec{\omega}$$

$\vec{r}$ is $\vec{r}_{/G}$

$$\underline{1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\vec{H}_{/G} = \underline{I}^G \cdot \vec{\omega}$$

$$\left[\int \vec{r}_{/G} \cdot \vec{r}_{/G} dm \right] \underline{1} - \int \vec{r}_{/G} \vec{r}_{/G} dm$$

$$\underline{I} = \int \underbrace{(\vec{r} \cdot \vec{r}) dm}_{|\vec{r}|^2} \underline{1} - \underbrace{\int (r_i \hat{e}_i)(r_j \hat{e}_j) dm}_{\left[\int r_i r_j dm \right] \hat{e}_i \hat{e}_j}$$

$$\Rightarrow \int \begin{bmatrix} x^2+y^2+z^2 & 0 & 0 \\ 0 & x^2+y^2+z^2 & 0 \\ 0 & 0 & x^2+y^2+z^2 \end{bmatrix} dm$$

As object moves around, I changes

$$\underline{I} = I_{ij} \hat{e}_i \hat{e}_j = \underline{I_{ij}'} \hat{e}_i' \hat{e}_j'$$

I_{ij}' at $t=0$

body fixed
constant in time

$$= \underline{I_{ij}^0} \hat{e}_i' \hat{e}_j'$$

$\uparrow$
 I^0

$$\hat{e}_i' = \underline{R} \cdot \hat{e}_i$$

$$\hat{e}_j = \hat{e}_j' \cdot \underline{R}^T$$

$$\boxed{\underline{I}(t) = \underline{R} \cdot \underline{I}^0 \cdot \underline{R}^T}$$

$$[\underline{I}(t)]_f = [\underline{R}]_f [\underline{I}^0]_f [\underline{R}]_f^T$$

$$\boxed{I = R I^0 R^T}$$

Back to AMB

$$\vec{M}_{/G} = \vec{H}_{/G}$$

$$\hookrightarrow \sum \vec{M}_{/G}^{\text{ext}} \hookrightarrow \frac{d}{dt} (\vec{H}_{/G})$$

$$\frac{d}{dt} (\underline{I} \cdot \vec{\omega}) = \underbrace{\frac{d}{dt} (\underline{I} \cdot \vec{\omega})}_{\text{"Q"}} + \vec{\omega} \times \underbrace{(\underline{I} \cdot \vec{\omega})}_{\text{"Q"}}$$

$$\frac{d \underline{I}}{dt} = \underline{0}$$

$$\frac{d}{dt} (\underline{I} \cdot \vec{\omega}) = \underline{I} \cdot \dot{\vec{\omega}} + \vec{\omega} \times (\underline{I} \cdot \vec{\omega})$$

$$\boxed{\frac{d}{dt} \vec{\omega} = \dot{\vec{\omega}}}$$

$$\Rightarrow \vec{M}_{/G} = \underline{I} \cdot \dot{\vec{\omega}} + \vec{\omega} \times (\underline{I} \cdot \vec{\omega})$$

$$\dot{\vec{\omega}} = \underline{I}^{-1} (\vec{M}_{/G} - \vec{\omega} \times (\underline{I} \cdot \vec{\omega}))$$

Matrix form:

$$\dot{\vec{\omega}} = \underline{I}^{-1} (\vec{M}_{/G} - \underbrace{\vec{\omega} \times (\underline{I} \cdot \vec{\omega})}_{3 \times 1})$$

$\begin{matrix} 3 \times 1 & 3 \times 3 & 3 \times 1 & 3 \times 1 & 3 \times 3 & 3 \times 1 \end{matrix}$

Have all we need to calculate motion of one rigid object

$$\boxed{\begin{aligned}\dot{\vec{v}}_G &= \frac{1}{m} \vec{F}^{\text{ext}} \\ \dot{\vec{r}}_G &= \vec{v}_G\end{aligned}}$$

not today's
lecture

$$\dot{\vec{\omega}} = \underline{\underline{I}}^{-1} (\vec{M}_{/G} - \vec{\omega} \times \underline{\underline{I}} \vec{\omega})$$

$$\underline{\underline{I}} = \underline{\underline{P}} \cdot \underline{\underline{I}}^0 \cdot \underline{\underline{P}}^T$$

$$\underline{\underline{P}} = \underline{\underline{\omega}} \cdot \underline{\underline{R}}$$

$$\underline{\underline{\omega}} = \underline{\underline{\Omega}} = \underline{\underline{\Omega}}(\alpha)$$

Initial value problem:

Give $\underline{\underline{P}}(0)$, $\underline{\underline{\omega}}(0)$, $\underline{\underline{I}}^0$] give all to ode45

Get back $\vec{\omega}(t)$, $\underline{\underline{P}}(t)$

Back to $\underline{\underline{I}}$

$$[\underline{\underline{I}}] = \int \begin{bmatrix} y^2 + z^2 & -xy & -xz \\ -xy & x^2 + z^2 & -yz \\ -xz & -yz & x^2 + y^2 \end{bmatrix} dm$$

$$\text{e.g.: } I_{11} = \int y^2 + z^2 dm$$

$$I_{12} = -\int xy dm$$